

LEARNING JAZZ PIANIST STYLE WITH CROSS-ATTENTION CONDITIONING

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ABSTRACT

Jazz pianists develop distinctive traits that experienced listeners can often identify within seconds, yet the features underlying this recognition resist formal description. We study jazz pianist style through the lens of a pretrained symbolic music transformer, showing that its learned representations already encode pianist identity well enough for highly accurate classification across two benchmarks. We then augment the transformer with cross-attention over learned pianist identity embeddings, enabling it to generate music conditioned on a specific artist’s style. Two evaluation protocols confirm that the generator captures meaningful stylistic structure: a sliding-window classifier consistently attributes conditioned continuations to the correct artist, far above unconditioned baselines; and a classifier trained *entirely* on synthetic generations identifies real pianists across 12 classes with 87% chunk-level and 95% song-level accuracy. Finally, we repurpose the classifier to locate the most characteristic moments within a performance, surfacing the specific musical gestures that distinguish each pianist’s voice.

1. INTRODUCTION

An experienced jazz listener can identify Art Tatum or Bill Evans within a few bars, yet the features that make their playing so recognizable resist precise description [1]. *DownBeat* magazine’s Blindfold Test—in which jazz musicians identify performers from recordings—has demonstrated this perceptual ability for over seven decades. Jazz pianists also directly imitate the greats in order to understand and incorporate their playing styles. Dick Hyman’s *In the Styles Of... The Great Jazz Pianists* [2] is a collection of original etudes, each written to embody a different pianist’s voice. Hyman describes Tatum’s “perfect legato” and “rapid runs in both hands,” Garner’s “solid touch” and “strumming, guitar-like left hand,” and Peterson’s “tremolos and glissandi” and “single-note lines interspersed with thirds.” His work poses a natural question for music gener-

ation research: can a model learn to internalize and reproduce these stylistic signatures?

Symbolic music generation systems can now produce convincing piano music [3,4], but conditioning on *individual performer identity*—rather than genre, era, or mood—remains largely unexplored. We study pianist style at three levels: embedding analysis reveals geometric structure in how pianists cluster (Section 3); classification confirms that style can be automatically distinguished (Section 4); and conditional generation tests whether it can be synthesized (Sections 5–7). A key challenge is that standard likelihood metrics are nearly blind to conditioning quality—models with and without artist conditioning produce near-identical perplexities (Table 2). This motivates classifier-based evaluation that directly probes whether generated music carries artist identity.

We make the following contributions. First, we show that an Aria-based [4] pianist classifier exceeds the state of the art on the Deep Pianist Identification benchmark [5] (Section 4). Second, we introduce a gated cross-attention conditional generator and evaluate it with two style-fidelity tests: a sliding-window agreement protocol showing a sustained ~ 33 -point advantage over baselines (Section 5), and a synthetic-transfer test [6] in which a 12-way classifier trained *only* on generated music reaches 95% song-level accuracy on real performances (Section 6). Third, we repurpose the classifier to locate the most stylistically distinctive moments within a performance (Section 7). Audio demos and more can be found at the project webpage.¹

2. RELATED WORK

Symbolic music generation. The Music Transformer [3] introduced relative self-attention, where attention weights depend on the distance between musical events rather than their absolute position, enabling the model to capture repetition and phrase structure. MuseNet [7] demonstrated composer-level conditioning by prepending genre, composer, and instrumentation tokens to the input sequence of a large-scale transformer. Oore et al. [8] pioneered expressive piano generation with an event-based MIDI representation encoding note onsets, offsets, velocities, and time shifts as discrete tokens—the same representational family that Aria builds on. More recently, MuseMorphose [9] applied a Transformer VAE to fine-grained pi-



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¹ <https://almostimplemented.github.io/jazz-pianist-style>

ano style transfer at the bar level, and Zhang et al. [10] conditioned symbolic generation on composer identity via VQ-diffusion. Aria [4] is a self-supervised piano transformer whose contrastive variant produces the per-track embeddings that we use for artist selection and classification. While score-to-performance rendering systems have conditioned on individual pianist identity to predict expressive parameters [11, 12], conditioning a generative model on performer identity to produce the musical content itself remains underexplored. Jazz makes this particularly compelling: unlike classical piano, where different performers interpret the same score and style resides primarily in expressive nuance, jazz pianists are simultaneously composers and performers. Their harmonic choices, melodic vocabulary, and rhythmic patterns are as much a part of their identity as their touch or phrasing. Modeling jazz pianist style therefore requires generating the musical content itself, not only rendering a given score expressively.

Performer identification. Cheston et al. [13] showed that rhythmic features extracted from source-separated audio—including swing ratio, tempo variability, and micro-timing deviations—predict jazz pianist identity well above chance; their subsequent Deep Pianist Identification benchmark [5] achieved 94.4% track-level accuracy with a ResNet-50 over piano-roll features. The PiJAMA dataset [14] provides aligned audio and MIDI for 30 jazz pianists. We build on both resources.

Cross-attention conditioning. Flamingo [15] introduced gated cross-attention layers for conditioning a frozen language model on a new modality; IP-Adapter [16] applied the pattern to image generation. We adopt the same gated-residual design but with a simpler conditioning source: a small learned per-artist embedding matrix injected into a subset of the model’s layers via cross-attention.

Evaluating conditional generation. Teacher-forced perplexity is standard but, as we show, nearly insensitive to conditioning quality. Distributional metrics such as Fréchet Audio Distance [17] measure aggregate similarity but do not test whether individual samples carry the intended conditioning. Classifier-based evaluation has been explored in vision [18]; the Classification Accuracy Score (CAS) [6]—training a classifier on generated data and testing on real data—directly inspires our synthetic transfer experiment (Section 6). High accuracy implies the generator has encoded class-discriminative structure that transfers to real examples.

3. DATA AND ARTIST SELECTION

We draw from PiJAMA-30 [14], a dataset of solo jazz piano performances spanning 30 artists with aligned audio and automatically transcribed MIDI. For this study, we select a subset of the 12 most distinctive pianists, which we call PiJAMA-12, focusing on artists whose styles are sufficiently differentiated for both a classifier and a conditional generator to exploit. To identify these artists, we compute per-track embeddings using the contrastive variant of Aria [4] and rank all 30 PiJAMA artists by k NN leave-

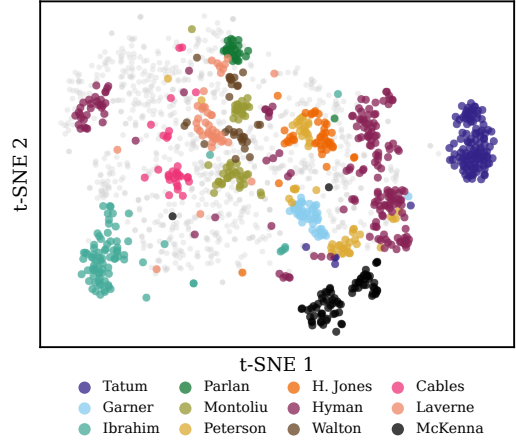


Figure 1. t-SNE [20] projection of Aria-CL track embeddings for all PiJAMA-30 pianists. The 12 selected pianists (colored) occupy distinct clusters; remaining artists are shown in light gray.

one-out accuracy and silhouette score [19], selecting the top 12. The selected pianists—Art Tatum, Erroll Garner, Abdullah Ibrahim, Horace Parlan, Tete Montoliu, Oscar Peterson, Hank Jones, Dick Hyman, Cedar Walton, George Cables, Andy Laverne, and Dave McKenna—occupy well-separated regions in embedding space (Figure 1), spanning a range of eras and stylistic approaches. This selection prioritizes stylistic separability over coverage; scaling to less separable artists is a natural next step.

After removing 22 PiJAMA-30 recordings that contain ensemble playing rather than solo piano, we tokenize all MIDI using Aria’s AbsTokenizer [4], with three tokens (pitch-velocity, onset, duration) per note, producing sequences of 4096 tokens (~ 1200 notes) for generation and 1024 tokens (~ 300 notes) for classification. All splits are at the song level to prevent leakage between train, validation, and test sets. The resulting PiJAMA-12 subset contains 779 songs (~ 57 hours) across the 12 artists, with per-artist counts ranging from 28 (Horace Parlan) to 187 (Dick Hyman), split into 1383 training, 175 validation, and 177 test sequences (at 4096 tokens; the classifier uses these same songs re-chunked into 1024-token windows).

4. APPROACH

4.1 Pianist Classification

We train a pianist classifier by fine-tuning the full Aria medium backbone (659M parameters; 16 layers, $d=1536$) with a linear classification head over the hidden state at the end-of-sequence token. Training uses class-balanced cross-entropy with pitch, tempo, and velocity augmentation [4] (learning rate 10^{-5} , batch size 16, 15 epochs).

To validate the feature backbone, we first evaluate on the Deep Pianist Identification benchmark [5] using their published splits and 20-class setup. Our classifier reaches 96.9% track-level accuracy, surpassing their ResNet-50 baseline of 94.4% (Table 1). We then train the classifier that serves as our evaluation tool throughout Sections 5

Method / Dataset	Chunk	Clip	Track
ResNet-50 [5] on DPI-20	—	80.5	94.4
Ours (Aria) on DPI-20	84.1	90.4	96.9
Ours on PiJAMA-12	95.8	—	98.8 [†]
Ours on PiJAMA-30	90.2	—	99.4 [†]

Table 1. Pianist identification accuracy (%). A *chunk* is a 1024-token classification window; a *clip* is a ~ 30 s audio segment following the protocol of [5]. Top: Deep Pianist benchmark (20 classes) using published splits; clip and track columns aggregate windows by mean logit. Bottom: PiJAMA classifiers (12 and 30 classes, identical recipe) on held-out song-level test splits (80 and 156 songs); the 12-class model is used for all style-fidelity evaluations. [†]Majority vote, ties broken by mean logit; each classifier misses one short track (of 80 and 156).

and 6 on the PiJAMA-12 subset, achieving 95.8% chunk-level and 98.8% track-level accuracy on the held-out test split. An identical recipe applied to all 30 PiJAMA artists reaches 90.2% and 99.4%, respectively (Table 1).

4.2 Cross-Attention Conditional Generation

Starting from the pretrained Aria medium model, we insert a gated cross-attention layer after the self-attention block in the last 8 of 16 transformer layers (Figure 2). Queries come from the music sequence; keys and values come from a learned artist embedding $E_{\text{artist}} \in \mathbb{R}^{4 \times d}$:

$$x' = x + g \cdot \text{CrossAttn}(x, E_{\text{artist}}) \quad (1)$$

where g is a scalar gate initialized to 0.1, following the Flamingo design [15]. This ensures the adapter starts as a near-identity residual. Unlike prompt-based conditioning, which loses influence as generation advances beyond the initial context [21], cross-attention provides a persistent channel that does not decay with sequence length. The 8 cross-attention layers and 12 artist embeddings add 78M parameters to the 659M base model (12% overhead). Element-wise dropout ($p=0.3$) is applied to the projected artist context during training as regularization.

We fine-tune the full model plus adapter for 15 epochs on PiJAMA-12 training sequences of 4096 tokens using AdamW (weight decay 0.02, 0.5-epoch linear warmup, effective batch size 32, peak learning rate 5×10^{-6} , fp16) on a single NVIDIA RTX 6000 GPU in approximately 18 hours. As an ablation control, we also train a baseline with identical hyperparameters but *no* cross-attention layers.

Table 2 reports token-weighted test-set perplexity. Fine-tuning closes nearly all of the pretrained-to-in-domain gap; cross-attention adds only a marginal further improvement. This is consistent with the well-known difficulty of training global conditioning signals in powerful autoregressive decoders [22]: under teacher forcing, the ground-truth prefix provides sufficient local context for accurate next-token prediction, leaving little room for a global signal such as pianist identity to improve the loss. The conditioning signal’s effect surfaces only during free generation, when the model must sustain

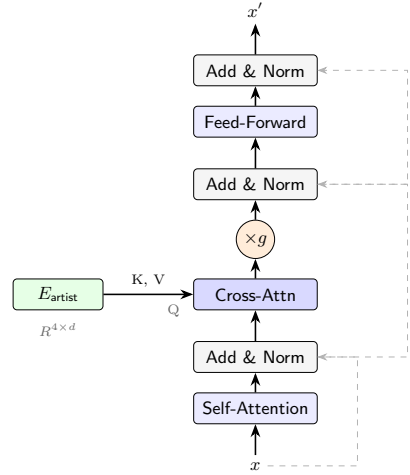


Figure 2. Gated cross-attention block added to the last 8 of 16 transformer layers. Queries come from the music sequence; keys and values from learned artist embeddings.

Model	Test NLL	Perplexity	Agreement [†]
Pretrained Aria	2.43	11.41	25%
Fine-tuned (no CA)	1.94	6.96	37%
Cross-attention (ours)	1.92	6.82	70%

Table 2. Token-weighted cross-entropy on the PiJAMA-12 test split (177 sequences, 562 125 tokens) and mean classifier agreement during free generation. Despite near-identical perplexities, style fidelity differs dramatically. [†]Mean sliding-window agreement at $P=256$ (Figure 3).

stylistic coherence from its own outputs. This motivates the classifier-based evaluations in the following sections.

5. STYLE PERSISTENCE EVALUATION

Since teacher-forced perplexity barely distinguishes our models, we introduce a classifier-based protocol that operates on the model’s own autoregressive output. For each validation sample with true artist A , we extract a prompt of length P tokens and generate a continuation of up to 4096 tokens. A 1024-token classification window is slid across the continuation at stride 128, and we record the fraction of windows where the classifier’s top-1 prediction matches A , as a function of window start position. We call this fraction the *agreement* with A (Table 2 reports its positional mean).

We evaluate four generation modes: (i) *conditioned* (cross-attention with true artist context), (ii) *ablation* (same architecture, conditioning context set to zero), (iii) *fine-tuned baseline* (no cross-attention layers), and (iv) *pretrained Aria* (no fine-tuning). All modes use temperature 0.95 with top- $k=50$ and top- $p=0.95$.

Figure 3 shows the results for $P \in \{128, 256, 512\}$ evaluated over the 175 validation sequences. The conditioned model maintains a large and stable advantage across the full 4096-token generation: at $P=256$, it averages 70% agreement with the true artist, versus 37% for the fine-tuned baseline and 25% for pretrained Aria. The ablation run—which uses the cross-attention architecture

but receives no conditioning context—tracks the fine-tuned baseline almost exactly, confirming that the gap is due to the conditioning signal, not the additional parameters. Notably, the conditioned curve decays at a similar rate to the baselines rather than remaining flat, suggesting the prompt provides complementary stylistic information that the conditioning signal does not fully replace. The conditioning signal’s influence is bounded by the learned gate g (Equation 1), which balances style fidelity against musical coherence: a gate too large would impose the artist embedding at the expense of continuity. The result is a stable *offset* in agreement rather than a different decay rate.

Figure 4 shows the per-artist breakdown at $P=256$. Agreement varies widely, from 96% (Hank Jones, Dick Hyman) to 29% (Cedar Walton). This variation does not simply reflect training data volume: Hank Jones achieves 96% from just 69 training sequences, while Dave McKenna reaches only 49% from nearly twice as many. Notably, Andy Laverne achieves high classification accuracy on real data (Figure 5) but only 35% conditioned agreement, indicating that a recognizable style does not guarantee the generator can reproduce it.

We also test a *mismatch* condition: prompting from artist A but conditioning on artist $B \neq A$. In individual generations, the classifier’s attribution shifts cleanly from the prompt artist to the conditioning artist partway through the continuation. In aggregate across 148 mismatch pairs at $P=512$, the prompt artist’s agreement decays from 40% to 27% while the conditioning artist’s rises from 15% to 24%, with the two converging near position ~ 1500 —confirming that the conditioning signal gradually overrides the prompt’s stylistic influence.

6. SYNTHETIC TRANSFER EVALUATION

The sliding-window test measures whether a *pretrained* classifier recognizes artist identity in generated output. A more demanding test removes any dependence on the pretrained classifier: we train a *new* classifier exclusively on generated MIDI, and evaluate it on real performances it has never seen.

Protocol. We take each PiJAMA-12 training sequence, use the first 256 tokens as a prompt, and generate a 4096-token continuation conditioned on the true artist. The prompt tokens are discarded; only the continuation is retained. These are re-chunked into 4782 sequences of 1024 tokens. An Aria-based classifier (same setup as Section 4.1) is trained exclusively on this synthetic data and evaluated on the real PiJAMA-12 test set.

Results. Table 3 shows that the synthetic-only classifier reaches 87.1% chunk-level and 95.0% song-level accuracy on real performances—within 9 chunk-level points of the real-data classifier—demonstrating that the generated MIDI encodes stylistic structure that transfers to real recordings. To isolate the classifier’s dependence on Aria pretraining, we repeat the experiment with a classifier that has no pretraining at all: a ResNet-50 over piano-roll clips, trained from random initialization with the architecture and recipe of [5] and model selection on held-out gener-

Training data	Chunk	Song [†]
Real PiJAMA-12	95.8	98.8
+ from scratch (ResNet-50)	93.1 [‡]	97.5
Synthetic only (ours)	87.1	95.0
+ from scratch (ResNet-50)	70.2 [‡]	91.3

Table 3. Pianist identification on the *real* PiJAMA-12 test set (80 songs, 591 chunks). Synthetic-trained classifiers see only generated continuations; within each architecture, the real- and synthetic-trained classifiers share one training recipe. [†]Majority vote over chunks within a song; exact vote ties are broken by mean logit. [‡]30 s piano-roll clips rather than token chunks.

ations. During training it never sees any MIDI data other than the output of our generative model—yet it identifies 91.3% of real test tracks (70.2% of 30 s clips; Table 3): recognizing the artist in the generated data requires no Aria-pretrained representations. Trained instead on the real training split, the same network reaches 97.5% of tracks (93.1% of clips); the synthetic-only ResNet is thus within 7 track-level points of its real-trained counterpart. Figure 5 breaks down chunk-level accuracy by artist: the gap between real and synthetic classifiers is largest for Cedar Walton and George Cables, who also have some of the lowest conditioned agreement (Figure 4).

Memorization and diversity. Three potential confounds deserve attention. First, since the generator was trained on the same data used as prompts, it might simply replay memorized sequences. Pitch and interval n -gram overlap between generated continuations and training data drops below 5% at $n=6$ and to 1% or below at $n=10$, and a stricter windowed piano-roll similarity [23, 24] agrees: generations match their closest training track *less* closely than held-out real performances do (mean max-Jaccard 0.20 vs. 0.32 over 8 s windows, excluding each continuation’s own source recording), and no generation is anomalously close to any single unrelated training or test track. Second, the generator might collapse to a single stereotyped output per artist. Pairwise 4-gram Jaccard similarity between same-artist generations is $1.5\text{--}3.5\times$ higher than between-artist generations—matching the ratio observed in real data—with absolute values far below 1.0. Third, Aria’s pretraining corpus may share source recordings with PiJAMA. Behaviorally, pretrained Aria shows no replay signature on our evaluation prompts—its unconditioned continuations collapse toward the corpus prior, not the true artist—and the transfer result survives with a classifier that has no pretraining at all; the two corpora were also transcribed by different systems, so even shared recordings would not yield identical token sequences.

7. CHARACTERISTIC REGION DETECTION

The evaluations in Sections 5 and 6 measure style fidelity in aggregate—across entire continuations or entire test sets. A complementary question is whether we can identify *where* in a performance the features that define a pianist’s style are most concentrated.

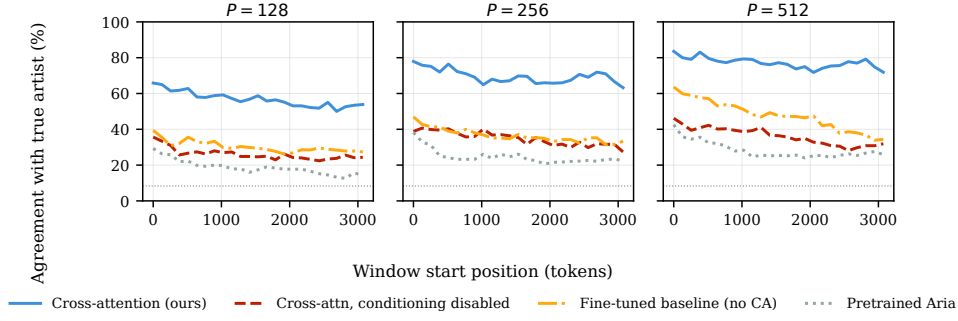


Figure 3. Classifier agreement with the true artist as a function of window-start position along a 4096-token continuation. Three panels show prompt lengths $P \in \{128, 256, 512\}$. The conditioned model (blue) sustains a ~ 33 -point advantage over the fine-tuned baseline across the full generation. The ablation (dashed red) tracks the fine-tuned baseline, isolating the effect of the conditioning signal from the architecture.

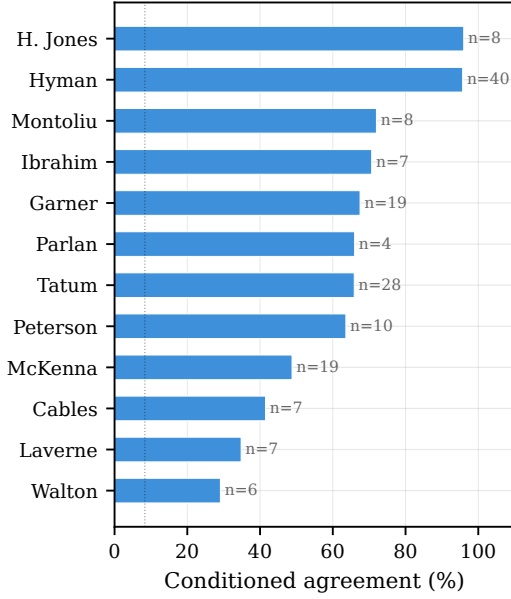


Figure 4. Per-artist conditioned agreement at $P=256$ (Section 5). Sample counts shown at right. Agreement varies from 96% (Hank Jones) to 29% (Cedar Walton); the dotted line marks chance (8.3%).

The classifier built in Section 4.1 assigns a single label to each chunk, but a performance is not uniformly characteristic: some passages may be more distinctively “Art Tatum” than others. By sliding the classifier across a held-out performance, we can attempt to locate these moments. A naïve approach would track the classifier’s raw confidence, but at 95.8% chunk-level accuracy the classifier is already near-certain on almost every window—raw logit margins hover around 15 ± 0.5 for a typical track, leaving almost no ranking signal. We therefore ask a relative question: where is the classifier *even more confident than usual* for this particular performance?

For each 1024-token window w at stride 128, we compute the *logit margin* between the true artist and the strongest alternative:

$$m(w) = \ell_{\text{true}}(w) - \max_{j \neq \text{true}} \ell_j(w) \quad (2)$$

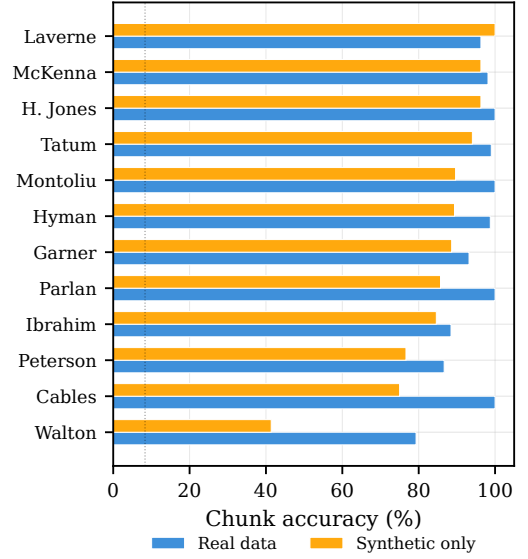


Figure 5. Per-pianist chunk-level test accuracy on PiJAMA-12. Blue: real-data classifier (Section 4.1). Orange: synthetic-only classifier (Section 6). Artists are ordered by synthetic accuracy.

We then normalize within each track:

$$z(w) = \frac{m(w) - \mu_T}{\sigma_T} \quad (3)$$

where μ_T and σ_T are the mean and standard deviation of m across all windows in track T . Intuitively, $z(w)$ measures how much more confidently the classifier recognizes the true pianist in window w compared to its average confidence across the whole performance. Gaussian smoothing ($\sigma=1.5$ windows) produces a continuous curve.

Figure 6 illustrates the result for Art Tatum’s “Sophisticated Lady.” The z-score curve (top) shows that most of the performance sits near zero—not because the classifier is uncertain, but because its high confidence is uniform. A peak near $t=125$ s stands out: the piano roll (bottom) shows this region contains dense note activity spanning over five octaves, consistent with the “rapid runs in both hands” that Hyman highlights in his Tatum etude (Section 1).

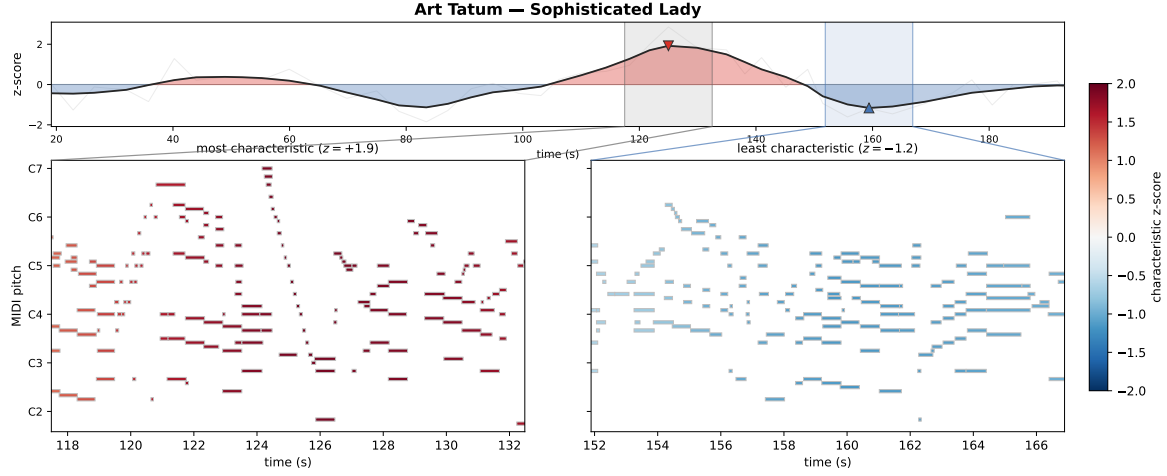


Figure 6. Characteristic region detection for Art Tatum, “Sophisticated Lady.” Top: within-track z-score of the classifier’s logit margin; red fill marks confidence above the track average, blue below. Bottom: piano rolls of the most (left, $z=+1.9$) and least (right, $z=-1.2$) characteristic 15-second regions, notes colored by z-score. The peak’s dense runs span over five octaves, consistent with Tatum’s virtuosic runs; the trough shows comparable note density (9.7 vs. 10.6 notes/s) over a narrower range (4.4 vs. 5.2 octaves), suggesting the classifier’s confidence tracks specific gestures, not overall activity.

Across all 80 test tracks (12 artists), the median peak z-score is 0.90 and the maximum is 1.99 (Art Tatum). The method reveals distinct patterns across artists. Artists like Art Tatum produce clear peaks, suggesting their style concentrates in specific gestures that the classifier finds especially discriminative. By contrast, Erroll Garner produces no comparable peaks across any of his four test tracks (maximum $z=0.59$), despite the classifier identifying him with equally high confidence on every window. For Garner, the features the classifier relies on appear to be distributed uniformly throughout a performance rather than concentrated in discrete moments.

8. DISCUSSION AND CONCLUSION

A recurring theme is the disconnect between token-level and sequence-level evaluation. Under teacher forcing, the ground-truth prefix provides rich local context that is highly predictive of the next token regardless of who is playing, leaving little room for a global signal like artist identity to improve the loss (Table 2). During free generation, however, small per-token biases accumulate: without conditioning, stylistic coherence degrades; with it, the artist embedding provides a persistent anchor. Pianist style is thus a *sequence-level* property, and our style-fidelity tests (Table 2, rightmost column) confirm that the conditioning signal carries information invisible to perplexity.

Some pianists consistently emerge as “easy” and “hard” across evaluations. Per Figure 4, Hank Jones and Dick Hyman achieve $>95\%$ conditioned agreement while Cedar Walton and Andy Laverne fall below 35%. Classification on real performances shows a similar pattern—Horace Parlan, who has the fewest training samples, achieves 100% chunk-level accuracy (Figure 5). These results suggest that *stylistic distinctiveness*, rather than data quantity, is the primary factor: pianists with a consistent and recognizable

voice are both easier to classify and to imitate.

The concentration of misclassifications on Dick Hyman is notable: for Art Tatum, 82% of incorrectly classified windows are attributed to Hyman; for Erroll Garner, 86%. This may reflect classifier bias toward the most-represented training class, or it may stem from Hyman’s mastery of various styles, making him a natural “default” for ambiguous passages. By contrast, George Cables’ errors distribute across multiple artists, suggesting a genuinely less distinctive generated style.

Several directions remain open. Combining artist conditioning with musical structure—lead sheets, chord progressions, or melodic constraints—would enable “in-the-style-of” generation over specified material. Mixing artist embeddings at inference could enable interpolation between styles, and classifier-free guidance [25] could strengthen the conditioning signal. A human listening study—ideally modeled on *DownBeat*’s Blindfold Test—would complement our classifier-based evaluations with perceptual judgments and remains the most important next step. Scaling beyond 12 artists—and beyond solo piano—would test the generality of the approach.

Taken together, our results show that a gated cross-attention adapter captures enough of each pianist’s style for a downstream classifier to learn from its output alone, and that the classifier itself can be repurposed to locate the musical gestures most characteristic of each artist. The characteristic-region method is particularly promising as a bridge between machine learning and musicology: it can surface aspects of style that are difficult to articulate formally, offering students and scholars a tool for understanding what makes each pianist’s voice distinctive. More broadly, these findings suggest that conditional generation models encode richer stylistic structure than standard evaluation reveals, and that classifier-based fidelity tests offer a practical means of accessing it.

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10. AI USAGE STATEMENT

An AI coding assistant (Anthropic Claude) was used throughout this work to help debug training code, generate figures, interface with the compute cluster, and write data-processing scripts. AI was also used as a writing assistant for L^AT_EX formatting, tabulation, literature review, and editing.

11. ETHICS STATEMENT

Style-conditioned music generation raises concerns about unauthorized imitation. AI-generated music imitating living artists has already appeared on streaming platforms without consent [26], causing real harm to musicians’ livelihoods and reputations. Our work is intended as a research tool for studying and understanding pianist style—a computational analogue of Dick Hyman’s pedagogical etudes [2], not a system for producing convincing forgeries. We encourage future work in this direction to consider consent, attribution, and the interests of the artists whose styles are modeled.

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